Anti-Money Laundering (AML) Risk Report

Executive Summary

Total Accounts Analysed:	**200**
High Risk Accounts (>=75):	**18**
Medium Risk Accounts (>=50):	**2**
Isolation Forest Anomalies:	**10**
Smurfing Flags:	**15**
Round-Tripping Flags:	**12**
Layering Flags:	**27**

This report was generated by an unsupervised machine-learning pipeline combining K-Means clustering, Isolation Forest anomaly detection, and rule-based AML typology detection (Smurfing, Round-Tripping, Layering). Accounts flagged as High Risk should be reviewed immediately by a certified AML analyst and escalated per regulatory requirements.

Top 10 High-Risk Accounts

Account	Score	Level	Smurf	RndTrip	Layer	Anomaly	Cluster
ACC2177	100.0	HIGH	Yes	No	Yes	Yes	1
ACC2095	96.8	HIGH	Yes	No	Yes	Yes	1
ACC2030	96.3	HIGH	Yes	No	Yes	Yes	1
ACC2128	93.1	HIGH	No	No	Yes	Yes	1
ACC2182	92.0	HIGH	Yes	No	Yes	Yes	1
ACC2068	89.3	HIGH	No	No	Yes	Yes	1
ACC2069	88.5	HIGH	Yes	No	Yes	Yes	1
ACC2115	85.8	HIGH	No	No	Yes	Yes	1
ACC2152	84.9	HIGH	Yes	No	Yes	No	1
ACC2066	83.5	HIGH	Yes	No	Yes	No	1

Suspicious Activity Reports (SAR) - Top 5 Accounts

SAR #1 | Account: ACC2177 | Score: 100.0 | HIGH

Smurfing Detected:	**Yes**
Round-Tripping Detected:	**No**
Layering Detected:	**Yes**
Isolation Forest Anomaly:	**Yes**
KMeans Cluster:	**1**
Avg Txn Amount (INR):	**6,716**
Txn Frequency / month:	**89**
Unique Counterparties:	**31**
Cash Deposit Ratio:	**0.94**
Intl Transfer Ratio:	**0.78**

Reason: High frequency + low amounts + high cash ratio -> possible SMURFING | Many counterparties + high international transfers -> possible LAYERING | Account belongs to highest-risk KMeans cluster | Statistically anomalous vs. peer group (Isolation Forest)

SAR #2 | Account: ACC2095 | Score: 96.8 | HIGH

Smurfing Detected:	**Yes**
Round-Tripping Detected:	**No**
Layering Detected:	**Yes**
Isolation Forest Anomaly:	**Yes**
KMeans Cluster:	**1**
Avg Txn Amount (INR):	**6,330**
Txn Frequency / month:	**86**
Unique Counterparties:	**21**
Cash Deposit Ratio:	**0.81**
Intl Transfer Ratio:	**0.74**

Reason: High frequency + low amounts + high cash ratio -> possible SMURFING | Many counterparties + high international transfers -> possible LAYERING | Account belongs to highest-risk KMeans cluster | Statistically anomalous vs. peer group (Isolation Forest)

SAR #3 | Account: ACC2030 | Score: 96.3 | HIGH

Smurfing Detected:	**Yes**
Round-Tripping Detected:	**No**
Layering Detected:	**Yes**
Isolation Forest Anomaly:	**Yes**
KMeans Cluster:	**1**
Avg Txn Amount (INR):	**5,628**
Txn Frequency / month:	**54**
Unique Counterparties:	**21**
Cash Deposit Ratio:	**0.95**
Intl Transfer Ratio:	**0.75**

Reason: High frequency + low amounts + high cash ratio -> possible SMURFING | Many counterparties + high international transfers -> possible LAYERING | Account belongs to highest-risk KMeans cluster | Statistically anomalous vs. peer group (Isolation Forest)

SAR #4 | Account: ACC2128 | Score: 93.1 | HIGH

Smurfing Detected:	**No**
Round-Tripping Detected:	**No**
Layering Detected:	**Yes**
Isolation Forest Anomaly:	**Yes**
KMeans Cluster:	**1**
Avg Txn Amount (INR):	**8,456**
Txn Frequency / month:	**77**
Unique Counterparties:	**32**
Cash Deposit Ratio:	**0.84**
Intl Transfer Ratio:	**0.63**

Reason: Many counterparties + high international transfers -> possible LAYERING | Account belongs to highest-risk KMeans cluster

SAR #5 | Account: ACC2182 | Score: 92.0 | HIGH

Smurfing Detected:	**Yes**
Round-Tripping Detected:	**No**
Layering Detected:	**Yes**
Isolation Forest Anomaly:	**Yes**
KMeans Cluster:	**1**
Avg Txn Amount (INR):	**5,617**
Txn Frequency / month:	**70**
Unique Counterparties:	**29**
Cash Deposit Ratio:	**0.87**
Intl Transfer Ratio:	**0.59**

Reason: High frequency + low amounts + high cash ratio -> possible SMURFING | Many counterparties + high international transfers -> possible LAYERING | Account belongs to highest-risk KMeans cluster

Risk Scoring Formula & Methodology

```
Risk_Score = (
    0.30 x Anomaly_Score_norm          (Isolation Forest - normalised [0,1])
  + 0.25 x Cash_Deposit_Ratio_norm     (min-max normalised)
  + 0.20 x Intl_Transfer_Ratio_norm    (min-max normalised)
  + 0.15 x Transaction_Frequency_norm  (min-max normalised)
  + 0.10 x Unique_Counterparties_norm  (min-max normalised)
) x 100
+ 10 x KMeans_Suspicious    (bonus: account in highest-risk cluster)
<- clipped to [0, 100]

Risk Level Thresholds:
  >= 75  ->  HIGH   (Mandatory SAR filing required)
  >= 50  ->  MEDIUM (Enhanced Due Diligence)
  <  50  ->  LOW    (Routine monitoring only)
```

Presentation Notes - Key Questions

1. Why unsupervised learning for AML?
   -> Labelled fraud data is scarce and biased; unsupervised methods find
   anomalies without needing prior fraud examples.

2. Legal obligation on detection?
   -> Banks MUST file a Suspicious Activity Report (SAR) with the national
   FIU (FinCEN/FIU-IND) within 30 days of detecting suspicious behaviour.
   Failure attracts heavy regulatory fines.

3. Reducing false positives in production?
   -> Ensemble the output of multiple models (K-Means + IF + Rules).
   -> Tune contamination rate with analyst feedback (active learning).
   -> Require ALL three methods to flag before escalating to HIGH.
   -> Incorporate customer segment peer-group comparison.